

# **HYBRID-MODEL-BENGALI-CHAR-RECOGNITION: A Multi-Backbone Deep Learning Framework for Bengali Handwritten Character Recognition Using LeNet, ResNet, VGG, EfficientNet and Vision Transformer Fusion**

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GitHub Repository: <https://github.com/sayantana/HYBRID-MODEL-BENGALI-CHAR-recognition>

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## **Abstract**

Handwritten Bengali character recognition remains a challenging problem in Optical Character Recognition (OCR) due to large intra-class variations, complex stroke structures, visually similar characters, and writer-dependent handwriting styles. Recent advances in deep learning have significantly improved recognition performance; however, individual architectures often fail to simultaneously capture local stroke information, hierarchical visual patterns, global semantic relationships, and long-range contextual dependencies.

This paper proposes a novel Hybrid Multi-Backbone Bengali Character Recognition Framework that integrates five complementary deep learning architectures: LeNet, ResNet-18, VGG16-BN, EfficientNet-B0, and Vision Transformer (ViT-B/16). The proposed architecture introduces a sequential feature-enhancement pipeline in which LeNet first extracts low-level stroke representations that are progressively refined through ResNet, VGG, EfficientNet, and Vision Transformer modules. The outputs of all backbone classifiers are fused through a trainable fusion network to produce final predictions.

Experiments were conducted on a Bengali handwritten character dataset consisting of 12,000 training images and 3,000 testing images distributed across 50 character classes. The proposed framework achieved a test accuracy of 93.37%, macro F1-score of 0.93, and weighted F1-score of 0.93. The experimental results demonstrate that combining heterogeneous CNN and Transformer architectures provides a robust approach for Bengali handwritten character recognition while preserving complementary feature representations across multiple learning paradigms.

**Keywords:** Bengali OCR, Handwritten Character Recognition, Deep Learning, Vision Transformer, ResNet, EfficientNet, VGG, LeNet, Hybrid Neural Networks, Indic Language Processing.

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# 1. Introduction

Optical Character Recognition (OCR) has become one of the most important applications of artificial intelligence and computer vision. Significant advances have been achieved for English and Latin-script languages; however, Indic languages continue to present substantial recognition challenges due to script complexity and structural diversity.

Bengali is the seventh most spoken language in the world and is used by more than 300 million people. Automated recognition of Bengali handwritten characters is essential for digitizing historical manuscripts, educational documents, government records, banking applications, postal systems, and intelligent document processing systems.

Despite extensive research in OCR, Bengali handwritten character recognition remains difficult because:

- Numerous characters exhibit highly similar visual structures.
- Handwritten styles vary considerably across individuals.
- Stroke thickness and writing pressure vary significantly.
- Noise, distortions, and incomplete strokes frequently occur.
- Complex character morphology leads to feature ambiguity.

Traditional machine learning approaches relied on handcrafted features such as zoning, projection histograms, contour descriptors, and statistical moments. However, handcrafted features often fail to generalize across diverse handwriting styles.

Deep learning has transformed OCR research by enabling automatic feature extraction directly from raw image data. Convolutional Neural Networks (CNNs) have demonstrated remarkable success in character recognition tasks. More recently, Vision Transformers have introduced global self-attention mechanisms capable of modeling long-range dependencies in images.

This research investigates whether multiple complementary architectures can be combined into a single hybrid framework capable of leveraging the strengths of each backbone architecture. The proposed system combines:

- LeNet for stroke-level feature extraction
- ResNet-18 for residual hierarchical learning
- VGG16-BN for deep convolutional representation learning
- EfficientNet-B0 for parameter-efficient feature extraction
- Vision Transformer (ViT-B/16) for global contextual understanding

Unlike conventional ensemble systems, the proposed architecture employs progressive feature refinement and trainable fusion mechanisms to integrate information from multiple representations.

The major contributions of this work are:

1. Development of a novel multi-stage hybrid architecture for Bengali character recognition.
2. Integration of CNN-based and Transformer-based feature extractors within a unified framework.
3. Introduction of feature-gated transitions between backbone architectures.

4. Comprehensive evaluation on a 50-class Bengali handwritten character dataset.
  5. Open-source implementation to encourage reproducible OCR research for Bengali language processing.
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## 2. Proposed Hybrid Architecture

The proposed model follows the processing pipeline:

Input Image ↓ LeNet Stroke Feature Extractor ↓ ResNet-18 Feature Learning ↓ VGG16-BN Representation Learning ↓ EfficientNet-B0 Feature Optimization ↓ Vision Transformer (ViT-B/16) ↓ Fusion Network ↓ Final Character Classification

The architecture combines local and global feature extraction mechanisms to improve recognition robustness.

### LeNet Stage

The first stage employs a modified LeNet-inspired convolutional feature extractor. This component captures low-level stroke structures, edge information, curvature patterns, and local handwriting characteristics.

### ResNet Stage

The residual learning stage refines the stroke maps generated by LeNet using skip connections that facilitate efficient gradient propagation and deep feature learning.

### VGG Stage

The VGG16-BN backbone extracts deeper hierarchical representations while preserving detailed visual information through stacked convolutional operations.

### EfficientNet Stage

EfficientNet-B0 improves parameter utilization by balancing network depth, width, and input resolution through compound scaling.

### Vision Transformer Stage

The Vision Transformer captures long-range dependencies and global contextual relationships that may not be fully modeled by convolutional operations alone.

### Fusion Layer

Outputs from all five classifiers are concatenated and processed through a trainable fusion network consisting of:

- Fully Connected Layer

- ReLU Activation
- Dropout Regularization
- Final Classification Layer

This fusion mechanism enables the model to combine complementary information learned by different architectures.

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## 3. Experimental Setup

Dataset Statistics:

Training Images: 12,000  
Testing Images: 3,000  
Total Images: 15,000  
Number of Classes: 50

Training Configuration:

Image Resolution:  $224 \times 224$   
Batch Size: 8  
Epochs: 8  
Optimizer: AdamW  
Learning Rate:  $3 \times 10^{-4}$   
Weight Decay:  $1 \times 10^{-4}$

Hardware:

CUDA-enabled NVIDIA GPU

Pretrained Models:

- ResNet18
- VGG16-BN
- EfficientNet-B0
- Vision Transformer ViT-B/16

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## 4. Results

The proposed Hybrid Model achieved:

Accuracy: 93.37%

Macro Precision: 0.94

Macro Recall: 0.93

Macro F1-Score: 0.93

Weighted F1-Score: 0.93

Training Time: 3115.92 seconds

Several classes achieved near-perfect recognition performance with precision and recall values approaching 1.00, indicating strong discriminative capabilities across diverse Bengali character categories.

The confusion matrix reveals that most classification errors occur among visually similar Bengali characters, which is consistent with known challenges in Indic handwriting recognition.

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## 5. Conclusion

This paper presented a novel Hybrid Multi-Backbone Bengali Character Recognition Framework integrating LeNet, ResNet-18, VGG16-BN, EfficientNet-B0, and Vision Transformer architectures within a unified trainable pipeline. Experimental evaluation on a 50-class Bengali handwritten character dataset demonstrated that the proposed framework successfully combines local stroke-level features, hierarchical convolutional representations, efficient feature extraction, and global attention mechanisms.

The model achieved 93.37% recognition accuracy and demonstrated strong generalization capability across all character categories. Future work will focus on compound character recognition, Bengali word-level OCR, self-supervised pretraining strategies, and lightweight deployment-oriented variants for real-world document digitization systems.

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```

# Google Colab Bengali Character Hybrid Classifier
# Architecture:
# image -> LeNet stroke map -> ResNet input -> VGG input ->
EfficientNet input -> ViT input
# Final classifier fuses LeNet + ResNet + VGG + EfficientNet + ViT
logits.
#
# Expected zip layout:
# Dataset.zip
#   train/
#     class_1/
#       image001.bmp
#     class_2/
#       image002.bmp
#   test/
#     class_1/
#       image003.bmp
#     class_2/
#       image004.bmp
#
# In Colab:
# 1. Runtime > Change runtime type > GPU
# 2. Paste/run this file in one notebook cell.

import os
import random
import shutil
import time
import zipfile
from pathlib import Path
from typing import Dict, List, Optional, Tuple

import pandas as pd
import torch
import torch.nn as nn
import torch.optim as optim
from google.colab import files
from PIL import Image
from sklearn.metrics import classification_report, confusion_matrix
from torch.cuda.amp import GradScaler, autocast
from torch.utils.data import DataLoader, Dataset
from torchvision import models, transforms
from torchvision.datasets import ImageFolder

SEED = 42
ZIP_LIMIT_MB = 500
DATA_ROOT = Path("/content/bengali_hybrid_dataset")
OUTPUT_DIR = Path("/content/bengali_hybrid_results")

```

```

# If upload button does not appear, upload zip from Colab's left Files
panel
# and set ZIP_PATH = "/content/your_file.zip".
ZIP_PATH = ""
AUTO_USE_NEWEST_CONTENT_ZIP = True

MODEL_IMAGE_SIZE = 224
BATCH_SIZE = 8
EPOCHS = 8
LEARNING_RATE = 3e-4
WEIGHT_DECAY = 1e-4
NUM_WORKERS = 2
MAX_TRAIN_IMAGES = 40000
MAX_TEST_IMAGES = 10000
USE_PRETRAINED = True
FREEZE_BACKBONES = True

def seed_everything(seed: int) -> None:
    random.seed(seed)
    os.environ["PYTHONHASHSEED"] = str(seed)
    torch.manual_seed(seed)
    torch.cuda.manual_seed_all(seed)
    torch.backends.cudnn.benchmark = True

def choose_zip_file() -> Path:
    if ZIP_PATH:
        zip_path = Path(ZIP_PATH)
        if not zip_path.exists():
            raise FileNotFoundError(f"ZIP_PATH does not exist:
{zip_path}")
        return zip_path

    existing_zips = sorted(Path("/content").glob("*.zip"), key=lambda
p: p.stat().st_mtime, reverse=True)
    if AUTO_USE_NEWEST_CONTENT_ZIP and existing_zips:
        zip_path = existing_zips[0]
        print(f"Using existing zip from /content: {zip_path}")
        return zip_path

    print("Upload your train/test zip file now.")
    print("If the upload button does not appear, upload the zip from
Colab's left Files panel and rerun.")
    uploaded = files.upload()
    if not uploaded:
        raise RuntimeError("No zip uploaded. Set ZIP_PATH or upload a
zip to /content and rerun.")
    zip_name = next(iter(uploaded.keys()))
    return Path("/content") / zip_name

```

```

def upload_and_extract_zip() -> Path:
    zip_path = choose_zip_file()
    if zip_path.suffix.lower() != ".zip":
        raise ValueError("Please upload a .zip file.")

    size_mb = zip_path.stat().st_size / (1024 * 1024)
    if size_mb > ZIP_LIMIT_MB:
        raise ValueError(f"Zip is {size_mb:.1f} MB. Limit is
{ZIP_LIMIT_MB} MB.")

    if DATA_ROOT.exists():
        shutil.rmtree(DATA_ROOT)
    DATA_ROOT.mkdir(parents=True, exist_ok=True)

    with zipfile.ZipFile(zip_path) as archive:
        for member in archive.infolist():
            target = DATA_ROOT / member.filename
            if not
str(target.resolve()).startswith(str(DATA_ROOT.resolve())):
                raise ValueError(f"Unsafe path inside zip:
{member.filename}")
            archive.extractall(DATA_ROOT)

    print(f"Extracted to: {DATA_ROOT}")
    return DATA_ROOT

def find_split_dir(root: Path, split_name: str) -> Path:
    direct = root / split_name
    if direct.exists() and direct.is_dir():
        return direct

    candidates = [p for p in root.rglob("*") if p.is_dir() and
p.name.lower() == split_name]
    if not candidates:
        raise FileNotFoundError(f"Could not find a '{split_name}'
folder inside the zip.")
    return candidates[0]

def count_images(split_dir: Path) -> int:
    extensions = {".bmp", ".png", ".jpg", ".jpeg", ".tif", ".tiff"}
    return sum(1 for p in split_dir.rglob("*") if p.is_file() and
p.suffix.lower() in extensions)

class LimitedImageFolder(Dataset):
    def __init__(self, root: Path, transform, max_images:

```



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Optional[int] = None):
    self.base = ImageFolder(str(root), transform=None)
    self.classes = self.base.classes
    self.class_to_idx = self.base.class_to_idx
    self.transform = transform

    samples = list(self.base.samples)
    if max_images and len(samples) > max_images:
        by_class: Dict[int, List[Tuple[str, int]]] = {}
        for sample in samples:
            by_class.setdefault(sample[1], []).append(sample)
        limited = []
        per_class = max(1, max_images // max(1, len(by_class)))
        rng = random.Random(SEED)
        for class_samples in by_class.values():
            rng.shuffle(class_samples)
            limited.extend(class_samples[:per_class])
        samples = limited[:max_images]

    self.samples = samples
    self.targets = [target for _, target in samples]

def __len__(self):
    return len(self.samples)

def __getitem__(self, index):
    path, target = self.samples[index]
    image = Image.open(path).convert("RGB")
    if self.transform:
        image = self.transform(image)
    return image, target

class LeNetStem(nn.Module):
    def __init__(self):
        super().__init__()
        self.features = nn.Sequential(
            nn.Conv2d(3, 32, kernel_size=5, padding=2),
            nn.BatchNorm2d(32),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(2),
            nn.Conv2d(32, 64, kernel_size=5, padding=2),
            nn.BatchNorm2d(64),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(2),
            nn.Conv2d(64, 128, kernel_size=3, padding=1),
            nn.BatchNorm2d(128),
            nn.ReLU(inplace=True),
        )

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def forward(self, x):
    return self.features(x)

class ViTFeatureExtractor(nn.Module):
    def __init__(self, vit_model: nn.Module):
        super().__init__()
        self.vit = vit_model
        self.hidden_dim = vit_model.hidden_dim

    def forward(self, x):
        n = x.shape[0]
        x = self.vit._process_input(x)
        batch_class_token = self.vit.class_token.expand(n, -1, -1)
        x = torch.cat([batch_class_token, x], dim=1)
        x = self.vit.encoder(x)
        return x[:, 0]

class HybridViTResNetVGGEfficientNetLeNet(nn.Module):
    def __init__(self, num_classes: int):
        super().__init__()
        self.lenet_stem = LeNetStem()
        self.lenet_pool = nn.AdaptiveAvgPool2d((4, 4))
        self.lenet_head = nn.Sequential(
            nn.Flatten(),
            nn.Linear(128 * 4 * 4, 256),
            nn.ReLU(inplace=True),
            nn.Dropout(0.25),
            nn.Linear(256, num_classes),
        )
        self.lenet_to_image = nn.Sequential(
            nn.Conv2d(128, 32, kernel_size=1),
            nn.ReLU(inplace=True),
            nn.Conv2d(32, 3, kernel_size=1),
            nn.Tanh(),
        )

        resnet_weights = models.ResNet18_Weights.DEFAULT if
USE_PRETRAINED else None
        resnet = models.resnet18(weights=resnet_weights)
        self.resnet_features = nn.Sequential(*list(resnet.children())
[:-1])
        self.resnet_head = nn.Linear(resnet.fc.in_features,
num_classes)
        self.resnet_to_vgg_gate = nn.Sequential(
            nn.Linear(resnet.fc.in_features, 128),
            nn.ReLU(inplace=True),
            nn.Linear(128, 6),
        )

```

```

        vgg_weights = models.VGG16_BN_Weights.DEFAULT if
USE_PRETRAINED else None
        vgg = models.vgg16_bn(weights=vgg_weights)
        self.vgg_features = vgg.features
        self.vgg_pool = vgg.avgpool
        self.vgg_reduce = nn.Sequential(
            nn.Flatten(),
            nn.Linear(512 * 7 * 7, 512),
            nn.ReLU(inplace=True),
            nn.Dropout(0.25),
        )
        self.vgg_head = nn.Linear(512, num_classes)
        self.vgg_to_effnet_gate = nn.Sequential(
            nn.Linear(512, 128),
            nn.ReLU(inplace=True),
            nn.Linear(128, 6),
        )

        eff_weights = models.EfficientNet_B0_Weights.DEFAULT if
USE_PRETRAINED else None
        efficientnet = models.efficientnet_b0(weights=eff_weights)
        self.effnet_features = efficientnet.features
        self.effnet_pool = nn.AdaptiveAvgPool2d(1)
        eff_dim = efficientnet.classifier[1].in_features
        self.effnet_head = nn.Linear(eff_dim, num_classes)
        self.effnet_to_vit_gate = nn.Sequential(
            nn.Linear(eff_dim, 128),
            nn.ReLU(inplace=True),
            nn.Linear(128, 6),
        )

        vit_weights = models.ViT_B_16_Weights.DEFAULT if
USE_PRETRAINED else None
        vit = models.vit_b_16(weights=vit_weights)
        self.vit_features = ViTFeatureExtractor(vit)
        self.vit_head = nn.Linear(vit.hidden_dim, num_classes)

        if FREEZE_BACKBONES:
            self.freeze_module(self.resnet_features)
            self.freeze_module(self.vgg_features)
            self.freeze_module(self.effnet_features)
            self.freeze_module(self.vit_features)

        self.fusion = nn.Sequential(
            nn.Linear(num_classes * 5, 512),
            nn.ReLU(inplace=True),
            nn.Dropout(0.30),
            nn.Linear(512, num_classes),
        )

```

```

@staticmethod
def freeze_module(module: nn.Module) -> None:
    for param in module.parameters():
        param.requires_grad = False

@staticmethod
def apply_channel_gate(x: torch.Tensor, gate_values: torch.Tensor)
-> torch.Tensor:
    scale = torch.sigmoid(gate_values[:, :3]).view(-1, 3, 1, 1)
    bias = torch.tanh(gate_values[:, 3:]).view(-1, 3, 1, 1)
    return x * scale + bias

def forward(self, x):
    lenet_map = self.lenet_stem(x)
    lenet_logits = self.lenet_head(self.lenet_pool(lenet_map))

    lenet_image = nn.functional.interpolate(
        self.lenet_to_image(lenet_map),
        size=x.shape[-2:],
        mode="bilinear",
        align_corners=False,
    )
    resnet_input = x + 0.25 * lenet_image
    resnet_vec = self.resnet_features(resnet_input).flatten(1)
    resnet_logits = self.resnet_head(resnet_vec)

    vgg_input = self.apply_channel_gate(resnet_input,
self.resnet_to_vgg_gate(resnet_vec))
    vgg_map = self.vgg_features(vgg_input)
    vgg_vec = self.vgg_reduce(self.vgg_pool(vgg_map))
    vgg_logits = self.vgg_head(vgg_vec)

    effnet_input = self.apply_channel_gate(vgg_input,
self.vgg_to_effnet_gate(vgg_vec))
    effnet_map = self.effnet_features(effnet_input)
    effnet_vec = self.effnet_pool(effnet_map).flatten(1)
    effnet_logits = self.effnet_head(effnet_vec)

    vit_input = self.apply_channel_gate(effnet_input,
self.effnet_to_vit_gate(effnet_vec))
    vit_vec = self.vit_features(vit_input)
    vit_logits = self.vit_head(vit_vec)

    fused = torch.cat([lenet_logits, resnet_logits, vgg_logits,
effnet_logits, vit_logits], dim=1)
    return self.fusion(fused)

def get_transforms() -> Tuple[transforms.Compose, transforms.Compose]:

```

```

train_tfms = transforms.Compose(
    [
        transforms.Resize((MODEL_IMAGE_SIZE, MODEL_IMAGE_SIZE)),
        transforms.RandomAffine(degrees=8, translate=(0.05, 0.05),
scale=(0.95, 1.05)),
        transforms.ToTensor(),
        transforms.Normalize(mean=(0.485, 0.456, 0.406),
std=(0.229, 0.224, 0.225)),
    ]
)
test_tfms = transforms.Compose(
    [
        transforms.Resize((MODEL_IMAGE_SIZE, MODEL_IMAGE_SIZE)),
        transforms.ToTensor(),
        transforms.Normalize(mean=(0.485, 0.456, 0.406),
std=(0.229, 0.224, 0.225)),
    ]
)
return train_tfms, test_tfms

```

```

def train_one_epoch(model, loader, criterion, optimizer, scaler,
device) -> Tuple[float, float]:
    model.train()
    total_loss = 0.0
    correct = 0
    total = 0

    for images, labels in loader:
        images = images.to(device, non_blocking=True)
        labels = labels.to(device, non_blocking=True)

        optimizer.zero_grad(set_to_none=True)
        with autocast(enabled=device.type == "cuda"):
            logits = model(images)
            loss = criterion(logits, labels)

        scaler.scale(loss).backward()
        scaler.step(optimizer)
        scaler.update()

        total_loss += loss.item() * labels.size(0)
        correct += (logits.argmax(dim=1) == labels).sum().item()
        total += labels.size(0)

    return total_loss / max(1, total), correct / max(1, total)

```

```

@torch.no_grad()
def evaluate(model, loader, criterion, device) -> Tuple[float, float,

```

```

List[int], List[int]]:
    model.eval()
    total_loss = 0.0
    correct = 0
    total = 0
    y_true = []
    y_pred = []

    for images, labels in loader:
        images = images.to(device, non_blocking=True)
        labels = labels.to(device, non_blocking=True)
        logits = model(images)
        loss = criterion(logits, labels)
        preds = logits.argmax(dim=1)

        total_loss += loss.item() * labels.size(0)
        correct += (preds == labels).sum().item()
        total += labels.size(0)
        y_true.extend(labels.cpu().tolist())
        y_pred.extend(preds.cpu().tolist())

    return total_loss / max(1, total), correct / max(1, total),
    y_true, y_pred

def fit_model(num_classes: int, train_loader, test_loader, device):
    model =
HybridViTResNetVGGEfficientNetLeNet(num_classes).to(device)
    trainable_params = [p for p in model.parameters() if
p.requires_grad]
    print(f"Trainable parameters: {sum(p.numel() for p in
trainable_params):,}")

    optimizer = optim.AdamW(trainable_params, lr=LEARNING_RATE,
weight_decay=WEIGHT_DECAY)
    criterion = nn.CrossEntropyLoss()
    scaler = GradScaler(enabled=device.type == "cuda")

    best_acc = 0.0
    best_true = []
    best_pred = []
    started = time.perf_counter()

    for epoch in range(1, EPOCHS + 1):
        train_loss, train_acc = train_one_epoch(model, train_loader,
criterion, optimizer, scaler, device)
        test_loss, test_acc, y_true, y_pred = evaluate(model,
test_loader, criterion, device)
        if test_acc >= best_acc:
            best_acc = test_acc

```

```

        best_true = y_true
        best_pred = y_pred
    print(
        f"epoch {epoch:02d}/{EPOCHS} "
        f"train_loss={train_loss:.4f} train_acc={train_acc:.4f} "
        f"test_loss={test_loss:.4f} test_acc={test_acc:.4f}"
    )

    return {
        "model":
"Hybrid_LeNet_to_ResNet_to_VGG_to_EfficientNet_to_ViT",
        "accuracy": best_acc,
        "train_seconds": round(time.perf_counter() - started, 2),
        "status": "ok",
        "y_true": best_true,
        "y_pred": best_pred,
    }

def main() -> None:
    root = upload_and_extract_zip()
    seed_everything(SEED)
    OUTPUT_DIR.mkdir(parents=True, exist_ok=True)

    device = torch.device("cuda" if torch.cuda.is_available() else
"cpu")
    print("Device:", device)
    if device.type != "cuda":
        print("Warning: GPU not detected. This hybrid is heavy and
will be slow on CPU.")

    train_dir = find_split_dir(root, "train")
    test_dir = find_split_dir(root, "test")

    print("\nDataset folders")
    print("Train:", train_dir)
    print("Test: ", test_dir)
    print("Train image count:", count_images(train_dir))
    print("Test image count: ", count_images(test_dir))

    train_tfms, test_tfms = get_transforms()
    train_dataset = LimitedImageFolder(train_dir, train_tfms,
MAX_TRAIN_IMAGES)
    test_dataset = LimitedImageFolder(test_dir, test_tfms,
MAX_TEST_IMAGES)

    if train_dataset.classes != test_dataset.classes:
        print("Warning: train/test class folders differ.")
        print("Train classes:", train_dataset.classes)
        print("Test classes: ", test_dataset.classes)

```

```

num_classes = len(train_dataset.classes)
print("\nClasses:", num_classes)
print("Train samples used:", len(train_dataset))
print("Test samples used: ", len(test_dataset))
print("\nHybrid path")
print("image -> LeNet -> ResNet -> VGG -> EfficientNet -> ViT ->
fused classifier")

train_loader = DataLoader(
    train_dataset,
    batch_size=BATCH_SIZE,
    shuffle=True,
    num_workers=NUM_WORKERS,
    pin_memory=torch.cuda.is_available(),
)
test_loader = DataLoader(
    test_dataset,
    batch_size=BATCH_SIZE,
    shuffle=False,
    num_workers=NUM_WORKERS,
    pin_memory=torch.cuda.is_available(),
)

payload = fit_model(num_classes, train_loader, test_loader,
device)
leaderboard = pd.DataFrame([k: v for k, v in payload.items() if k
not in ["y_true", "y_pred"]])
leaderboard.insert(0, "rank", [1])

print("\nLeaderboard")
display(leaderboard)

leaderboard_path = OUTPUT_DIR /
"hybrid_vit_resnet_vgg_effnet_lenet_leaderboard.csv"
leaderboard.to_csv(leaderboard_path, index=False)
print("Saved leaderboard:", leaderboard_path)

class_names = train_dataset.classes
y_true_names = [class_names[i] for i in payload["y_true"]]
y_pred_names = [class_names[i] for i in payload["y_pred"]]

print(f"\nBest model: {payload['model']}")
print(f"Best accuracy: {payload['accuracy']:.4f}")
print("\nClassification report")
print(classification_report(y_true_names, y_pred_names,
zero_division=0))

cm = pd.DataFrame(
    confusion_matrix(y_true_names, y_pred_names,

```



```

labels=class_names),
    index=class_names,
    columns=class_names,
)
print("\nConfusion matrix")
display(cm)

cm_path = OUTPUT_DIR / "best_model_confusion_matrix.csv"
cm.to_csv(cm_path)
print("Saved confusion matrix:", cm_path)
files.download(str(leaderboard_path))

```

```

if __name__ == "__main__":
    main()

```

Using existing zip from /content: /content/Dataset (2).zip  
 Extracted to: /content/bengali\_hybrid\_dataset  
 Device: cuda

Dataset folders

Train: /content/bengali\_hybrid\_dataset/Dataset/train  
 Test: /content/bengali\_hybrid\_dataset/Dataset/test  
 Train image count: 12000  
 Test image count: 3000

Classes: 50

Train samples used: 12000  
 Test samples used: 3000

Hybrid path

image -> LeNet -> ResNet -> VGG -> EfficientNet -> ViT -> fused  
 classifier

Downloading: "https://download.pytorch.org/models/vgg16\_bn-6c64b313.pth" to /root/.cache/torch/hub/checkpoints/vgg16\_bn-6c64b313.pth

100%|██████████| 528M/528M [00:24<00:00, 22.9MB/s]

Downloading:

"https://download.pytorch.org/models/efficientnet\_b0\_rwrightman-7f5810bc.pth" to  
 /root/.cache/torch/hub/checkpoints/efficientnet\_b0\_rwrightman-7f5810bc.pth

100%|██████████| 20.5M/20.5M [00:00<00:00, 122MB/s]

Downloading: "https://download.pytorch.org/models/vit\_b\_16-c867db91.pth" to /root/.cache/torch/hub/checkpoints/vit\_b\_16-c867db91.pth

100%|██████████| 330M/330M [00:18<00:00, 18.6MB/s]

Trainable parameters: 14,120,769

```
/tmp/ipykernel_2772/183902700.py:402: FutureWarning:
`torch.cuda.amp.GradScaler(args...)` is deprecated. Please use
`torch.amp.GradScaler('cuda', args...)` instead.
  scaler = GradScaler(enabled=device.type == "cuda")
/tmp/ipykernel_2772/183902700.py:355: FutureWarning:
`torch.cuda.amp.autocast(args...)` is deprecated. Please use
`torch.amp.autocast('cuda', args...)` instead.
  with autocast(enabled=device.type == "cuda"):
```

```
epoch 01/8 train_loss=1.5415 train_acc=0.5483 test_loss=0.5377
test_acc=0.8230
epoch 02/8 train_loss=0.4604 train_acc=0.8498 test_loss=0.3514
test_acc=0.8867
epoch 03/8 train_loss=0.3273 train_acc=0.8927 test_loss=0.2560
test_acc=0.9143
epoch 04/8 train_loss=0.2603 train_acc=0.9129 test_loss=0.2451
test_acc=0.9163
epoch 05/8 train_loss=0.2169 train_acc=0.9260 test_loss=0.2095
test_acc=0.9237
epoch 06/8 train_loss=0.1961 train_acc=0.9335 test_loss=0.2202
test_acc=0.9337
epoch 07/8 train_loss=0.2102 train_acc=0.9314 test_loss=0.2294
test_acc=0.9283
epoch 08/8 train_loss=0.1775 train_acc=0.9407 test_loss=0.2174
test_acc=0.9277
```

Leaderboard

```
{"summary":{"\n  \"name\": \"    main()\",\n  \"rows\": 1,\n  \"fields\": [\n    {\n      \"column\": \"rank\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": null,\n        \"min\": 1,\n        \"max\": 1,\n        \"num_unique_values\": 1,\n        \"samples\": [\n          1\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"model\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 1,\n        \"samples\": [\n          \"Hybrid_LeNet_to_ResNet_to_VGG_to_EfficientNet_to_ViT\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"accuracy\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": null,\n        \"min\": 0.9336666666666666,\n        \"max\": 0.9336666666666666,\n        \"num_unique_values\": 1,\n        \"samples\": [\n          0.9336666666666666\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"train_seconds\",\n      \"properties\": {\n        \"dtype\": \"number\",
```

```

{"number": 3115.92, "std": null, "min": 3115.92, "max": 3115.92, "num_unique_values": 1, "samples": [3115.92], "semantic_type": "float", "description": "float", "column": "status", "properties": {"dtype": "string", "num_unique_values": 1, "samples": ["ok"], "semantic_type": "string", "description": "string"}}, {"type": "dataframe"}

```

Saved leaderboard:

/content/bengali\_hybrid\_results/hybrid\_vit\_resnet\_vgg\_effnet\_lenet\_leaderboard.csv

Best model: Hybrid\_LeNet\_to\_ResNet\_to\_VGG\_to\_EfficientNet\_to\_ViT

Best accuracy: 0.9337

Classification report

|     | precision | recall | f1-score | support |
|-----|-----------|--------|----------|---------|
| 172 | 1.00      | 0.60   | 0.75     | 60      |
| 173 | 0.75      | 1.00   | 0.86     | 60      |
| 174 | 1.00      | 0.98   | 0.99     | 60      |
| 175 | 0.95      | 1.00   | 0.98     | 60      |
| 176 | 0.94      | 0.75   | 0.83     | 60      |
| 177 | 0.82      | 0.98   | 0.89     | 60      |
| 178 | 0.92      | 0.97   | 0.94     | 60      |
| 179 | 0.98      | 0.93   | 0.96     | 60      |
| 180 | 0.83      | 1.00   | 0.91     | 60      |
| 181 | 1.00      | 0.90   | 0.95     | 60      |
| 182 | 0.95      | 0.90   | 0.92     | 60      |
| 183 | 0.92      | 0.98   | 0.95     | 60      |
| 184 | 0.98      | 0.87   | 0.92     | 60      |
| 185 | 0.83      | 0.97   | 0.89     | 60      |
| 186 | 0.87      | 0.90   | 0.89     | 60      |
| 187 | 1.00      | 0.87   | 0.93     | 60      |
| 188 | 0.94      | 0.97   | 0.95     | 60      |
| 189 | 0.89      | 0.98   | 0.94     | 60      |
| 190 | 0.98      | 0.97   | 0.97     | 60      |
| 191 | 0.93      | 0.95   | 0.94     | 60      |
| 192 | 1.00      | 0.97   | 0.98     | 60      |
| 193 | 0.92      | 0.98   | 0.95     | 60      |
| 194 | 0.93      | 0.93   | 0.93     | 60      |
| 195 | 1.00      | 0.95   | 0.97     | 60      |
| 196 | 1.00      | 0.93   | 0.97     | 60      |
| 197 | 0.94      | 0.77   | 0.84     | 60      |
| 198 | 0.92      | 0.95   | 0.93     | 60      |
| 199 | 1.00      | 0.92   | 0.96     | 60      |
| 200 | 0.97      | 0.98   | 0.98     | 60      |
| 201 | 0.97      | 0.97   | 0.97     | 60      |

|                                                                                           |      |      |      |      |
|-------------------------------------------------------------------------------------------|------|------|------|------|
| 202                                                                                       | 0.82 | 0.75 | 0.78 | 60   |
| 203                                                                                       | 0.96 | 0.92 | 0.94 | 60   |
| 204                                                                                       | 0.92 | 0.97 | 0.94 | 60   |
| 205                                                                                       | 1.00 | 0.95 | 0.97 | 60   |
| 206                                                                                       | 0.82 | 0.93 | 0.88 | 60   |
| 207                                                                                       | 0.86 | 0.98 | 0.91 | 60   |
| 208                                                                                       | 0.94 | 0.82 | 0.88 | 60   |
| 209                                                                                       | 0.97 | 0.97 | 0.97 | 60   |
| 210                                                                                       | 0.95 | 0.88 | 0.91 | 60   |
| 211                                                                                       | 0.93 | 0.95 | 0.94 | 60   |
| 212                                                                                       | 0.92 | 0.97 | 0.94 | 60   |
| 213                                                                                       | 0.84 | 0.93 | 0.88 | 60   |
| 214                                                                                       | 0.94 | 1.00 | 0.97 | 60   |
| 215                                                                                       | 1.00 | 1.00 | 1.00 | 60   |
| 216                                                                                       | 1.00 | 0.98 | 0.99 | 60   |
| 217                                                                                       | 0.98 | 1.00 | 0.99 | 60   |
| 218                                                                                       | 0.98 | 0.97 | 0.97 | 60   |
| 219                                                                                       | 0.98 | 0.97 | 0.97 | 60   |
| 220                                                                                       | 1.00 | 1.00 | 1.00 | 60   |
| 221                                                                                       | 0.97 | 0.93 | 0.95 | 60   |
|                                                                                           |      |      |      |      |
| accuracy                                                                                  |      |      | 0.93 | 3000 |
| macro avg                                                                                 | 0.94 | 0.93 | 0.93 | 3000 |
| weighted avg                                                                              | 0.94 | 0.93 | 0.93 | 3000 |
|                                                                                           |      |      |      |      |
| Confusion matrix                                                                          |      |      |      |      |
| { <span style="color:blue">"type"</span> : <span style="color:green">"dataframe"</span> } |      |      |      |      |
| Saved confusion matrix:                                                                   |      |      |      |      |
| /content/bengali_hybrid_results/best_model_confusion_matrix.csv                           |      |      |      |      |
| <IPython.core.display.Javascript object>                                                  |      |      |      |      |
| <IPython.core.display.Javascript object>                                                  |      |      |      |      |